

Complete Cloud Computing Conversation Notes

AWS S3 (Simple Storage Service)

AWS S3 is an object storage service from AWS.
It stores data as objects inside buckets.

Main Features:

- Highly scalable
- 99.999999999% durability
- Lifecycle management
- Versioning
- Encryption
- Static website hosting

Workflow:

User → Upload Object → S3 Bucket → Storage

Use Cases:

- Backup
 - Media storage
 - Big data
 - AI datasets
 - Static websites
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Azure Blob Storage

Azure Blob Storage is Microsoft Azure's object storage service.

Blob Types:

- Block Blob
- Append Blob
- Page Blob

Storage Tiers:

- Hot
- Cool
- Archive

Architecture:

Application → Azure Blob Storage → Persistent Cloud Storage

Use Cases:

- Video storage
 - Image hosting
 - Backup
 - Analytics
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AWS EC2

AWS EC2 stands for Elastic Compute Cloud.

It provides virtual machines in the cloud.

Features:

- Auto Scaling
- Elastic IP
- Security Groups
- Load Balancing
- Multiple Instance Types

Architecture:

User → EC2 Instance → Operating System → Application

Use Cases:

- Web hosting
 - AI/ML
 - Big data
 - Enterprise apps
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Azure Virtual Machines

Azure Virtual Machines provide scalable virtual servers.

Supports:

- Windows
- Linux

Features:

- VM Scale Sets
- Virtual Networking
- Managed Disks
- Availability Zones

Workflow:

User Request → Azure VM → OS → Application Execution

AWS SageMaker

AWS SageMaker is a managed machine learning platform.

Capabilities:

- Data preparation
- Training
- Hyperparameter tuning
- Deployment
- Monitoring

Components:

- SageMaker Studio
- Autopilot
- Feature Store
- Pipelines

ML Workflow:

Data → Training → Model → Endpoint → Predictions

Azure Machine Learning

Azure Machine Learning is Microsoft's AI/ML platform.

Features:

- Experiment tracking
- MLOps
- Automated ML
- Pipelines
- Model Registry

Workflow:

Dataset → Training → Evaluation → Deployment

Vertex AI

Vertex AI is Google Cloud's end-to-end AI platform.

Features:

- AutoML
- Generative AI
- Pipelines
- TPU Support
- Feature Store
- Matching Engine

Architecture:

Data → Vertex AI → Training → Deployment → Monitoring

Supports:

- Gemini models
 - LLMs
 - RAG systems
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Docker

Docker is a containerization platform.

Core Concepts:

- Docker Image
- Docker Container
- Dockerfile
- Docker Hub

Workflow:

Dockerfile → Build Image → Run Container

Advantages:

- Lightweight
- Portable
- Fast deployment
- Isolation

Example Dockerfile:

```
FROM python:3.10
WORKDIR /app
COPY . .
RUN pip install flask
CMD ["python", "app.py"]
```

Orchestration

Orchestration means automated coordination of systems.

Functions:

- Scaling
- Scheduling
- Networking
- Monitoring
- Self-healing

Popular Tools:

- Kubernetes
- Docker Swarm
- Apache Mesos

Workflow:

Containers → Orchestrator → Deployment + Scaling

Kubernetes

Kubernetes is a container orchestration platform.

Main Components:

- Pods
- Nodes
- Deployments
- Services
- Ingress
- ConfigMaps
- Secrets

Architecture:

Control Plane → Worker Nodes → Pods

Features:

- Auto scaling
- Rolling updates
- Self-healing
- Service discovery

Deployment Flow:

YAML → API Server → Scheduler → Pod Deployment

Cloud-Native

Cloud-Native applications are designed specifically for cloud environments.

Main Pillars:

- Containers
- Microservices
- Orchestration
- DevOps
- CI/CD

Architecture:

Microservices → Containers → Kubernetes → Cloud Infrastructure

Benefits:

- Scalability
 - Resilience
 - Faster deployment
 - Fault tolerance
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AWS Lambda

AWS Lambda is a serverless computing service.

Characteristics:

- Event-driven
- Auto scaling
- Stateless
- Pay-per-use

Triggers:

- S3
- API Gateway
- DynamoDB
- EventBridge

Workflow:

Event → Lambda Function → Execute → Response

Important Concepts:

- Cold Start
 - Warm Start
 - Firecracker MicroVM
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Azure Functions

Azure Functions is Microsoft's serverless compute platform.

Features:

- Trigger-based execution
- Bindings
- Durable Functions
- Auto scaling

Triggers:

- HTTP Trigger
- Timer Trigger
- Blob Trigger
- Queue Trigger

Workflow:

Event → Azure Function → Execution → Response

Use Cases:

- APIs
 - Automation
 - File processing
 - AI workflows
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Important Cloud Computing Concepts

Concept	Meaning
Containers	Lightweight isolated runtime environments
Microservices	Independent modular services
Serverless	Run code without server management
CI/CD	Continuous Integration and Deployment
Auto Scaling	Dynamic resource scaling
MLOps	Machine Learning Operations
IaC	Infrastructure as Code
Pod	Smallest deployable Kubernetes unit
Orchestration	Automated coordination of systems

End of Complete Conversation Notes